Relative Polygonalization with Prescribed Traces for Convex Neural Codes

Alec Kriebel

Independent Researcher

me@aleckriebel.com | ORCID: 0009-0001-9320-500X

Version 1.0 — August 2026

Abstract

Bukh and Jeffs proved that every open convex neural code realizable in the plane admits a realization by interiors of convex polygons. We establish a relative form of their theorem in which finitely many affine-line traces and finitely many atom witnesses are fixed in advance. Given a compact convex polygon F, open convex fields $U_1, \ldots, U_n$, and finitely many lines, we construct convex polygons whose interiors realize exactly the same code in $\operatorname{int} F$, retain every protected witness, and agree with the source fields on every prescribed line after clipping by F. The construction may simultaneously retain the associated closed line sections, which handles tangent contacts, point traces, half-open clipped traces, simultaneous endpoints, and lower-dimensional atoms. As consequences, two planar models with common source data can be made to agree pointwise on a shared edge, and the four faces of a source tetrahedron admit a coherent polygonal boundary model preserving all six edge traces. The result supplies exact planar interface data for higher-dimensional constructions, but it does not provide compatible normal derivatives or a convex three-dimensional filling.

Keywords. convex neural code; polygonal realization; prescribed line trace; convex Venn diagram; interface model; place field.
2020 MSC. 52C99 (primary); 52A10, 05B30, 92B20 (secondary).

1 Introduction

A neural code records which members of a finite family of receptive fields are active at each point of an ambient space. When the fields are required to be open and convex, the resulting combinatorial objects link convex geometry, combinatorics, topology, and mathematical neuroscience (Curto et al., 2013, 2017; Cruz et al., 2019). Even in the plane, a realization may begin with curved, tangent, or degenerate boundaries, so a basic structural question is whether the same code can be represented by polygons.

Bukh and Jeffs (2023) answered this affirmatively: every planar open convex code has a realization by interiors of convex polygons. Their proof first passes to a suitable closed realization, then chooses an inclusion-minimal subrealization and proves that every member of such a minimal tuple is polygonal. The theorem is exact at the level of the code, but its statement does not require a specified cross-section or a specified atom witness to remain fixed.

The purpose of this paper is to show that the planar construction has an exact relative form. Finitely many line sections can be frozen before the minimal shrinking step, and finitely many selected atom witnesses can be protected. The key finite datum on a prescribed line is not merely the order of endpoints after clipping by the ambient polygon. One records the complete bounded line section first. A four-point trace frame forces every point of its open section to remain interior after shrinking, while the closed endpoints prevent movement of the section. Only afterward is the line section clipped by the ambient polygon. This order of operations is what preserves half-open traces at ambient vertices and simultaneous endpoint events.

Our main result is stated in theorem 5.2. In addition to the open trace identity requested there, the proof preserves the complete closed section of every prescribed line after an auxiliary bounded reduction. This slightly stronger formulation makes tangent contacts and point sections transparent. Protected witnesses are retained by small triangles placed inside their active fields.

The relative theorem has two immediate interface applications. First, if two planar face problems share an edge and inherit the same source trace, their polygonal outputs agree pointwise in strict membership on that edge. Second, all six edge traces of a source tetrahedron can be prescribed simultaneously on its four triangular faces, producing a coherent finite polygonal model of the complete boundary sphere. These statements concern membership traces only. They do not match numerical defining functions, normal slopes, or three-dimensional convex fillings.

Relationship to prior work. The baseline polygonalization theorem and the local simplification mechanism are due to Bukh and Jeffs (2023). The new ingredient is a finite trace frame and the resulting relative bookkeeping: arbitrary finitely many line sections, protected atom witnesses, a fixed ambient polygon, tangent closed contacts, compatible shared-edge outputs, and the tetrahedral-boundary application. Standard approximation of convex bodies by polytopes concerns metric error rather than exact Boolean atoms and exact prescribed sections; see, for example, Schneider and Wieacker (1981). Exact interpolation also appears in the opposite smoothing direction—for example, prescribed subsets of polytope facets can be retained while smoothing a single body (Ghomi, 2004)—but that problem does not preserve a multi-set neural atom code. Our argument is instead an exact constrained shrinking argument for an entire convex family.

Organization. After definitions in section 3, section 4 reviews the inclusion-minimal planar construction in the form needed here. Section 5 builds the finite trace data and proves the main theorem. Section 7 treats tangent, clipped, and lower-dimensional cases. Section 8 gives shared-edge and finite facewise compatibility, and section 9 proves the tetrahedral-boundary application. Section 10 records the precise scope and the absence of a three-dimensional filling theorem.

2 The relative strengthening in context

The main theorem is not a new proof that planar convex codes are polygonal; that result is due to Bukh and Jeffs (2023). Its contribution is to make a finite collection of geometric data survive the same shrinking process. The distinction is summarized in table 1.

The proof does not approximate the fields in Hausdorff distance. It shrinks the closed tuple while preserving a finite representative package. The trace frames, ambient vertices, and protected triangles become additional immutable data in the inclusion-minimal construction. This is why exact atoms and exact line sections can be controlled simultaneously even in the presence of tangencies and closure-supported behavior.

Feature	Planar polygonalization	Present relative theorem
Carrier	An open planar realization	A fixed compact polygon and its relative interior
Exact neural code	Preserved	Preserved relative to the fixed carrier
Prescribed lines	Used internally for closure-only patterns	Arbitrary finite lines added to the internal family
Strict traces	No external trace constraint in the theorem statement	Exact source subsets retained after clipping
Closed contacts	Used as proof data where needed	Any prescribed point or segment section may be retained
Atom witnesses	A representative of each codeword is used	Arbitrarily designated witnesses remain at their original coordinates
Interface consequence	Not part of the theorem statement	Shared edges and finite facewise models agree pointwise
Higher-dimensional filling	Not addressed	Explicitly not claimed

Table 1: Comparison with the baseline planar theorem. The present result strengthens its finite constraints, not its planar dimension.

3 Preliminaries

For $n \geq 1$, write $[n] = \{1, \ldots, n\}$. Let $X \subseteq \mathbb{R}^d$ and let $\mathcal{U} = (U_1, \ldots, U_n)$ be a finite family of subsets of X. For $x \in X$, its membership pattern is

$$\mathrm{patt}_{\mathcal{U}}(x) = \{i \in [n] : x \in U_i\}.$$

The associated code is

$$\mathrm{code}(\mathcal{U}; X) = \{\mathrm{patt}_{\mathcal{U}}(x) : x \in X\} \subseteq 2^{[n]}.$$

Equivalently, a word $\sigma \subseteq [n]$ occurs when its atom

$$A_\sigma(\mathcal{U}; X) = X \cap \bigcap_{i \in \sigma} U_i \setminus \bigcup_{j \notin \sigma} U_j$$

is nonempty. We call the code *open convex* when X and all U_i are open and convex in a common finite-dimensional affine space.

Throughout the planar theorem, $F \subset \mathbb{R}^2$ is a compact convex polygon. The relative source code is

$$\mathrm{code}_F(\mathcal{U}) = \mathrm{code}(U_1 \cap \operatorname{int} F, \ldots, U_n \cap \operatorname{int} F; \operatorname{int} F).$$

A *protected witness* is a selected point $x \in \operatorname{int} F$ together with its source word $\mathrm{patt}_{\mathcal{U}}(x)$. Preserving it means that its membership word is unchanged in the output realization.

Let L be an affine line. The *open trace* of U_i on L relative to F is

$$L \cap U_i \cap F,$$

and the corresponding *closed trace* is $L \cap \overline{U_i} \cap F$. The ambient factor F is closed in these expressions. Thus an open source interval can become half-open or closed after clipping by F.

A compact convex polygon is permitted to be empty when the corresponding neuron is trivial. In the one- and zero-dimensional relative versions, "polygon" means respectively a compact interval or a point, with the same empty-set convention.

Remark 3.1 (Relative and ambient interiors). The output field associated with a two-dimensional polygon P_i is its Euclidean interior. On a segment or a point carrier, all interiors are understood in the relative affine hull. A closed tangent contact may be retained even though it does not belong to the strict open trace.

4 The inclusion-minimal planar construction

We recall the parts of the planar argument of Bukh and Jeffs (2023) that are needed for the relative theorem. The two points that matter are: a compact convex realization can be shrunk subject to finitely many retained incidences, and an inclusion-minimal planar realization is polygonal.

4.1 Constrained inclusion-minimal tuples

Let $\mathcal{X} = (X_0, \dots, X_n)$ be a tuple of compact convex subsets of $\mathbb{R}^2$, and let $R \subset \mathbb{R}^2$ be finite. We say that a subtuple $\mathcal{Y} = (Y_0, \dots, Y_n)$ is *R-admissible* when

$$Y_i \subseteq X_i, \qquad Y_i \cap R = X_i \cap R, \qquad \mathrm{code}(\mathcal{Y}) = \mathrm{code}(\mathcal{X}).$$

Lemma 4.1 (Constrained minimal tuple)**.** *Assume that R contains a representative of every word of* $\mathrm{code}(\mathcal{X})$*. Then every nonempty family of R-admissible tuples contains a coordinatewise inclusion-minimal member.*

Proof. Order the admissible tuples by coordinatewise inclusion. For a descending chain, take the coordinatewise intersection. Compactness and convexity are preserved, as are the incidences on R. Every source word still occurs, because R will contain a representative for every closed-code word. If a word is realized at a point of the chain intersection, then, for each coordinate absent at that point, choose a member of the chain that already excludes the point. One member of the totally ordered chain is contained in all finitely many chosen members and realizes the same word. Thus no new word appears in the intersection. Zorn's lemma gives a minimal member. ☐

4.2 Planar polygonality

The geometric result of Bukh and Jeffs (2023) is the following constrained form of their inclusion-minimality proposition.

Lemma 4.2 (Planar polygonality)**.** *Every member of an R-admissible inclusion-minimal tuple in the plane is a convex polygon, a segment, a point, or the empty set.*

Proof sketch and retained constraints. The proof is the local planar simplification argument of Bukh and Jeffs (2023). If one member has infinitely many extreme points, choose an extreme accumulation point outside the finite representative set. Inside a sufficiently small disk, every incident convex boundary meets the circle in one connected arc and no unrelated boundary or representative is encountered. Replace each incident set locally by the cone from the chosen point over its circular trace. Radial projection to the circle shows that no membership pattern changes, while at least one coordinate shrinks properly, contradicting inclusion minimality.

The argument tolerates finitely many additional retained points because the local disk is chosen disjoint from them. It also tolerates a coordinate that is a fixed polygon and contains all its vertices in R: a local simplification at a nonvertex boundary point does not change the convex hull of those vertices. Segments, points, and the empty set are already polytopal. We use no vertex-count estimate in this paper. ☐

4.3 Open codes from closed realizations

Let $\mathcal{U} = (U_0, \dots, U_n)$ be bounded open convex sets in the plane and put $X_i = \overline{U_i}$. Passing to closures may introduce codewords, but only on lower-dimensional active intersections.

Lemma 4.3 (Closure-only words)**.** *If $c \in \operatorname{code}(X_0, \dots, X_n) \setminus \operatorname{code}(U_0, \dots, U_n)$, then $\bigcap_{i \in c} X_i$ has empty Euclidean interior.*

Proof. Assume the active closed intersection has nonempty interior. Let q realize c in the closed tuple and choose p in the interior of the active intersection. The segment from p to q, except possibly at q, lies in that interior. Since q has positive distance from every inactive closed set, a nearby point q' on the segment still avoids all inactive sets. It lies in $\operatorname{int} X_i = U_i$ for all active i, and therefore realizes c in the open tuple, a contradiction. □

For every nonempty η such that

$$X_\eta := \bigcap_{i \in \eta} X_i$$

is nonempty and has empty interior, choose one affine line L_η containing X_η; convexity in the plane makes this possible. Denote the finite set of these lines by $\mathcal{L}_{\text{deg}}$.

Lemma 4.4 (Open-code recovery)**.** *Let $\mathcal{Y} = (Y_0, \dots, Y_n)$ be a compact convex subtuple of $\mathcal{X} = (X_0, \dots, X_n)$ with the same closed code. Suppose:*

1. *for every $L \in \mathcal{L}_{\text{deg}}$ and every i,*

$$L \cap \operatorname{int} Y_i = L \cap U_i;$$

2. *for every word $c \in \operatorname{code}(\mathcal{U})$, a retained triangle has vertices in Y_i for all $i \in c$ and contains a source point of word c in its interior.*

Then

$$\operatorname{code}(\operatorname{int} Y_0, \dots, \operatorname{int} Y_n) = \operatorname{code}(\mathcal{U}).$$

Proof. First prove the inclusion from the new open code to the source open code. On every line in $\mathcal{L}_{\text{deg}}$, the complete strict membership pattern is unchanged by hypothesis. Let $L = \bigcup_\eta L_\eta$, and take $p \notin L$. Put $c = \operatorname{patt}_{\mathcal{Y}}(p)$. Since $\operatorname{code}(\mathcal{Y}) = \operatorname{code}(\mathcal{X})$, this word belongs to the closed source code. By theorem 4.3, any closed-source word absent from the open source can occur only inside one of the active intersections used to define $\mathcal{L}_{\text{deg}}$, and hence only on L. Thus c is a source open word.

The retained triangle for c implies that $\bigcap_{i \in c} Y_i$ has nonempty interior; choose q in that interior. For sufficiently small $\varepsilon > 0$, set

$$p_\varepsilon = p + \varepsilon(p - q).$$

The perturbation continues the ray from the interior point q through p. If $p \in \operatorname{int} Y_i$, then p_ε remains in Y_i for all sufficiently small ε. If $p \in \partial Y_i$ and $q \in \operatorname{int} Y_i$, convexity implies that every sufficiently small point beyond p on this ray lies outside Y_i. Finally, if $p \notin Y_i$, nonmembership persists for small ε because Y_i is closed. Hence

$$\operatorname{patt}_{(\operatorname{int} Y_i)}(p) = \operatorname{patt}_{\mathcal{Y}}(p_\varepsilon).$$

Taking ε small also keeps $p_\varepsilon \notin L$, so the right-hand word is a source open word. This proves that no new open word is created away from L, and the trace hypothesis handles points on L.

Conversely, take $c \in \operatorname{code}(\mathcal{U})$ and its retained triangle. The source witness in the interior of that triangle lies in $\operatorname{int} Y_i$ for every $i \in c$. For $j \notin c$, the witness lies outside $U_j = \operatorname{int} X_j$; since $Y_j \subseteq X_j$, it cannot lie in $\operatorname{int} Y_j$. Hence the witness retains the exact word c. □

The preceding proof is the open-to-closed step of Bukh and Jeffs (2023), with the finite line traces and witness triangles made explicit. The prescribed lines in our theorem will simply be added to $\mathcal{L}_{\text{deg}}$.

5 Polygonalization with prescribed traces

We now add finitely many exact line constraints and protected atom witnesses to the inclusion-minimal construction.

5.1 Bounded sections and trace frames

Choose a bounded open convex disk W with $F \Subset W$, and replace each source field by $U_i \cap W$. This changes no membership datum in F. Put

$$X_i = \overline{U_i \cap W}.$$

For every prescribed line L, the closed section $L \cap X_i$ is a compact segment, a point, or the empty set. If the open section is nonempty, then

$$L \cap (U_i \cap W) = (a, b), \qquad L \cap X_i = [a, b]$$

in an affine coordinate on L.

Lemma 5.1 (Trace frame)**.** *Let $U \subset \mathbb{R}^2$ be bounded, open, and convex, put $X = \overline{U}$, and let L be a line.*

1. *If $L \cap U = (a, b)$ is nonempty, there is a convex quadrilateral $Q \subset X$ having $[a, b]$ as one diagonal.*
2. *If $L \cap U = \varnothing$ but $L \cap X$ is nonempty, retaining the extreme points of $L \cap X$ forces every convex set $Y \subseteq X$ containing those points to satisfy $L \cap Y = L \cap X$, while $L \cap \operatorname{int} Y = \varnothing$.*

Proof. In the first case, the midpoint m of $[a, b]$ belongs to U. For a unit normal ν to L and sufficiently small $\epsilon > 0$, the points $m \pm \epsilon\nu$ belong to U. The convex hull of $a, b, m + \epsilon\nu, m - \epsilon\nu$ is the required quadrilateral.

In the second case, $L \cap X$ is a compact segment or point. A convex subset of X containing its extreme points contains the whole section and cannot contain anything outside it. Since $U = \operatorname{int} X$ is disjoint from L, no point of the retained section can become an ambient interior point of $Y \subseteq X$. $\square$

The construction records the complete section of the bounded field on the whole line, before clipping by the ambient polygon. This order is essential when a trace endpoint coincides with an ambient vertex.

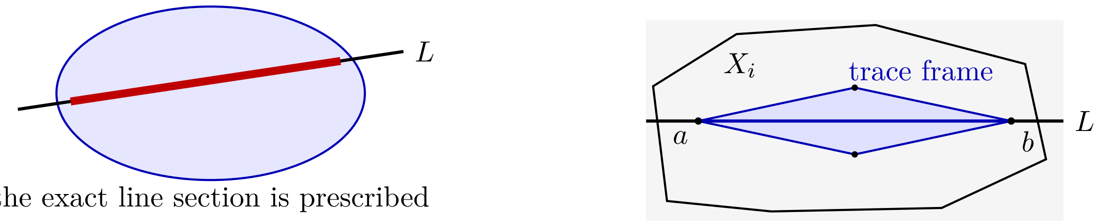
5.2 The finite representative package

Adjoin one ambient coordinate

$$U_0 = \operatorname{int} F, \qquad X_0 = F.$$

Let $\mathcal{L}$ be the union of the prescribed lines and the finite closure-degeneracy family $\mathcal{L}_{\text{deg}}$. Choose a finite set R containing:

1. one point of every atom of the closed tuple $(X_0, X_1, \ldots, X_n)$;



(a) A prescribed line and its full bounded section. (b) A four-point trace frame freezes the section.

Figure 1: The uncut section is retained before the ambient polygon is applied.

2. for every source open word, the vertices of a sufficiently small triangle lying in every active open field and containing a source witness of that word in its interior; every prescribed witness is used as the interior point of its triangle;
3. for every $L \in \mathcal{L}$ and every nonempty open section, all four vertices of a trace frame;
4. for every remaining nonempty closed section $L \cap X_i$, its extreme point or points;
5. every vertex of F.

All data are finite.

Theorem 5.2 (Finite-line relative polygonalization)**.** *Let $F \subset \mathbb{R}^2$ be a compact convex polygon with nonempty interior, let $U_1, \ldots, U_n$ be open convex sets, and let $L_1, \ldots, L_m$ be finitely many affine lines. Fix finitely many atom witnesses in* $\operatorname{int} F$*. Then there are compact convex polygons $P_1, \ldots, P_n$ such that, with $V_i = \operatorname{int} P_i$:*

1. *the exact relative code is preserved,*
$$\operatorname{code}(V_1 \cap \operatorname{int} F, \ldots, V_n \cap \operatorname{int} F; \operatorname{int} F) = \operatorname{code}_F(\mathcal{U});$$
2. *every prescribed strict trace is preserved,*
$$L_s \cap V_i \cap F = L_s \cap U_i \cap F;$$
3. *simultaneously, every prescribed closed section is preserved,*
$$L_s \cap P_i \cap F = L_s \cap \overline{U_i} \cap F;$$
4. *every protected witness remains in its original atom.*

The analogous statement holds in the relative affine hull when F is a segment or a point.

Proof. Apply theorem 4.1 to the bounded closed tuple $(X_0, X_1, \ldots, X_n)$ with the representative package R. Let $(Y_0, Y_1, \ldots, Y_n)$ be inclusion minimal. Since $Y_0 \subseteq F$ and contains every vertex of F, one has $Y_0 = F$. By theorem 4.2, all remaining Y_i are polygons, segments, points, or empty. Put $P_i = Y_i$ and $V_i = \operatorname{int} P_i$.

On every line in $\mathcal{L}_{\mathrm{deg}}$, the trace-frame and supporting section data give the exact strict source trace. The protected triangles remain. Hence theorem 4.4 gives equality of the complete open codes of
$$(\operatorname{int} F, V_1, \ldots, V_n) \quad \text{and} \quad (U_0, U_1 \cap W, \ldots, U_n \cap W).$$

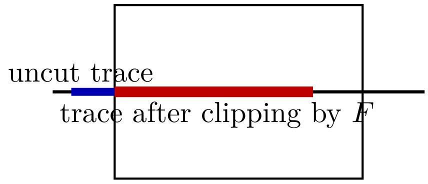


Figure 2: The same uncut open section can produce open, half-open, or closed relative traces after intersection with the closed ambient section.

Selecting the words containing the ambient coordinate gives the relative code equality. The same triangle argument preserves each protected witness exactly.

Fix a prescribed line $L = L_s$ and a neuron i. If the open bounded section is (a, b), the trace frame lies in P_i. Every point of (a, b) is interior to the quadrilateral and hence to P_i. The endpoints cannot be interior to $P_i \subseteq X_i$, and no point outside $[a, b]$ can enter P_i. Therefore

$$L \cap \operatorname{int} P_i = L \cap (U_i \cap W), \qquad L \cap P_i = L \cap X_i.$$

If the open section is empty, theorem 5.1 gives the same two identities. Since $F \Subset W$,

$$X_i \cap F = \overline{U_i} \cap F,$$

so clipping the identities by F proves both prescribed trace clauses for the original fields.

On a segment or point carrier, use the same construction in the relative affine hull. Inclusion-minimal compact convex sets there are intervals, points, or empty. □

Corollary 5.3 (Protected planar polygonalization)**.** *Every planar open convex code with finitely many designated atom witnesses has a polygonal realization retaining those witnesses.*

Proof. Choose a compact polygonal carrier containing the designated witnesses and one witness for every codeword, and apply theorem 5.2 with no prescribed lines. □

6 Exact trace configurations

The theorem is qualitative, but several elementary configurations explain why the complete section, the ambient coordinate, and closed supporting contacts must all be retained.

6.1 Clipping after recording the uncut section

Suppose a bounded field has line section (a, b) and the ambient polygon meets the same line in $[c, d]$. The relative trace is

$$(a, b) \cap [c, d].$$

Depending on the endpoint order, this set may be open, half-open, closed, or empty as a subset of the line. Freezing only the clipped order would lose the information that, for example, $c = a$ is a source endpoint rather than an arbitrary ambient endpoint. The trace frame freezes a and b first; the fixed ambient coordinate then performs the clipping.

6.2 Lower-dimensional atoms

Let P_1 be a rectangle, and let P_2 and P_3 be its upper and lower halves with their common horizontal side retained only in the closed model. Then the strict output code contains the word $\{1\}$ along that horizontal segment, while the adjacent open regions carry $\{1, 2\}$ and $\{1, 3\}$. This is an exact lower-dimensional atom. No positive-radius witness is available, but the open-to-closed argument retains the word through its line trace.

6.3 Simultaneous events

If two neurons have a common trace endpoint a on a prescribed line, the representative package contains the same geometric point a for both closed sections. The output therefore retains literal endpoint equality; it does not merely preserve the two endpoint orders separately. This is what is needed when two face models must agree on a shared edge.

7 Degenerate, tangent, and clipped traces

The stronger closed-section clause in theorem 5.2 makes the boundary cases explicit.

Corollary 7.1 (Tangent and lower-dimensional trace preservation)**.** *The construction preserves all of the following exactly.*

1. *If a prescribed line is tangent to a source neuron at a single closed point, the output has the same closed point contact and empty open trace.*
2. *If a supporting line meets a source closure in a nontrivial segment but misses the open field, the complete closed segment is retained and the strict trace remains empty.*
3. *If a source trace endpoint coincides with an ambient vertex, the clipped open trace may be half-open; its endpoint inclusion is unchanged.*
4. *Endpoint coincidences for several neurons remain simultaneous because every line section is retained as an actual subset, not merely through an order type.*
5. *Lower-dimensional atoms and closure-supported atoms remain codewords; any selected witness in such an atom is retained.*

Proof. The first two assertions follow from the exact closed- and open-section identities. The third follows because the complete bounded line section is frozen before intersecting with the closed ambient polygon. The fourth is pointwise equality of several retained traces. The last assertion follows from exact code equality and theorem 5.3. □

Remark 7.2 (Empty and whole traces)*.* An empty trace requires no frame. If the field contains the entire portion of a prescribed line that meets F, the bounded auxiliary disk turns the uncut trace into a compact interval that contains the clipped section in its interior; the same argument applies.

Corollary 7.3 (Covered relative carriers)**.** *Suppose a specified subfamily of the source neurons covers* $\operatorname{int} F$*. The corresponding output polygons cover* $\operatorname{int} F$ *as well.*

Proof. The relative empty word with respect to that subfamily is absent from the source code. Exact relative code equality prevents it from appearing in the output. No approximation or Lebesgue-number argument is needed. □

(a) A globally open interval becomes half-open after clipping by an ambient edge.

(b) A tangent line has empty strict trace and a retained closed point contact.

Figure 3: Boundary cases are controlled by recording complete line sections before ambient clipping.

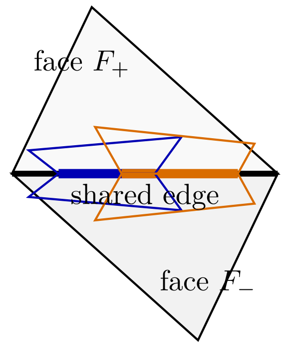


Figure 4: Two facewise polygonal models share the same strict membership intervals on their common edge. No equality of normal slopes is asserted.

8 Coherent planar interface models

Corollary 8.1 (Compatible shared-edge models)**.** *Let two planar convex-code problems have compact polygonal carriers F_- and F_+ sharing an edge e. Suppose that, for every neuron i, the two source fields induce the same strict trace on e. Apply theorem 5.2 to each face while prescribing the supporting line of e. Then the two polygonal outputs agree pointwise in strict membership on e. Their trace endpoints, tangent contacts, and simultaneous endpoint events are identical.*

Proof. Both applications preserve the same source subset $e \cap U_i$ exactly. The closed-section refinement likewise preserves the same closed source section when it is included in the data. □

The same observation gives a finite facewise statement.

Proposition 8.2 (Finite facewise-compatible models)**.** *Let K be a finite two-dimensional polyhedral complex whose facets are convex polygons in their affine planes. Suppose each facet carries a finite open convex source realization, and the source traces agree on every shared edge. Then each facet admits a polygonal realization with the same face code, and all incident realizations agree pointwise on every shared edge. Protected face witnesses and finitely many additional line traces may be retained.*

Proof. Apply theorem 5.2 independently to each facet, prescribing all of its edge-supporting lines and the additional lines. There are finitely many facets. Agreement on a shared edge follows from theorem 8.1. □

Remark 8.3 (No global convexity across the complex)*.* Theorem 8.2 is a coherent facewise output model. It does not claim that the union of the face pieces belonging to one neuron is convex in the ambient space, or even that it is the trace of one higher-dimensional convex set.

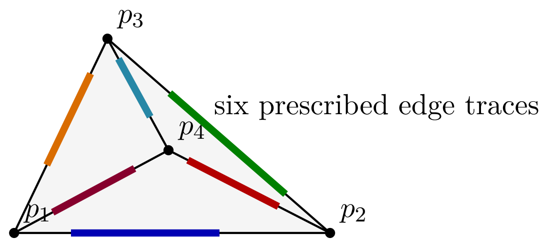


Figure 5: All six source edge traces are frozen first; the four faces are then polygonalized independently but coherently along the shared edges.

9 A coherent tetrahedral-boundary model

Let p_1, p_2, p_3, p_4 be affinely independent protected source witnesses in an open convex ambient set X, and put

$$T = \operatorname{conv}\{p_1, p_2, p_3, p_4\}.$$

After a small homothety about the barycenter if necessary, the four witnesses lie in a bounded relative carrier compactly contained in X.

For each neuron and each of the six edges $e_{rs} = [p_r, p_s]$, convexity gives a relatively open interval trace, possibly empty or the whole edge. Record all six traces in one common affine parametrization before treating the faces.

Theorem 9.1 (Coherent tetrahedral-boundary model)**.** *The four triangular faces of T admit polygonal neural-code models such that:*

1. *the code on each face equals the source code restricted to the relative interior of that face and is therefore a subcode of the global source code;*
2. *every source word occurring on a face survives;*
3. *every strict trace on each of the six edges is unchanged;*
4. *the two incident face models agree pointwise on every shared edge, including endpoint positions and simultaneous events.*

Thus ∂T carries a coherent finite facewise polygonal output model.

Proof. Apply theorem 5.2 to each triangular face in its relative affine plane, prescribing its three edge-supporting lines and protecting the desired face witnesses. A shared edge carries one source subset for every neuron, and both incident applications preserve it exactly. The statements follow. $\square$

Remark 9.2 (What is not supplied). The theorem does not construct convex three-dimensional polytopes whose boundary traces are the four face models. In particular, it does not match numerical gauge values, supporting facets, or inward normal derivatives across an edge. Those are genuinely additional three-dimensional compatibility conditions.

10 Scope, field of definition, and limitations

The theorem is an exact planar statement. It preserves full Boolean atom data in a polygonal carrier and finitely many one-dimensional sections. It does not address polygonal approximation in a metric norm, optimize the number of vertices, or provide a higher-dimensional filling.

Field of definition. Prescribed trace locations are not moved. Rational finite data lead to a first-order feasibility problem with rational coefficients and therefore to an algebraic output, but the proof does not guarantee rational vertices in general. Algebraic data yield algebraic output. If a prescribed endpoint is transcendental, it must remain so; the construction is interpreted in the real closure of the ordered field generated by the retained source data. No universal rationalization claim is made.

Relation to the higher-dimensional conjecture. Polytope-convex neural codes are characterized by representable oriented matroids (Kunin et al., 2023). It remains unknown whether every open convex code in higher dimension is polytope convex. The present result provides exact face and interface data useful in attempts at higher-dimensional gluing, but it does not settle that conjecture.

Normal data. Pointwise equality of edge membership traces is weaker than equality of finite concave defining functions on the edge, and much weaker than compatibility of their normal slopes. The tetrahedral-boundary theorem must not be read as a convex filling theorem.

Acknowledgments and declarations

The author thanks the authors of the planar polygonalization theorem for the proof framework on which the relative construction is based.

AI-assisted work. OpenAI ChatGPT, model GPT-5.6 Pro, was used extensively for exploratory proof reconstruction, drafting, exact-example code, literature searches, and adversarial editorial review. Several stronger higher-dimensional claims were rejected during that process and are not included here. The author reviewed the final manuscript and verification output and assumes responsibility for every mathematical claim, citation, and conclusion. The AI system is not an author.

Code availability. A deterministic no-network verification package accompanies the manuscript. It checks the finite rational examples, prescribed traces, protected witnesses, shared-edge data, tetrahedral edge consistency, manifests, and fresh compilation. The computation is illustrative and does not replace the general proof.

Funding and competing interests. This research received no specific grant. The author declares no competing interests.

A Trace bookkeeping in affine coordinates

Fix an affine coordinate s on a prescribed line L. After the bounded reduction, the closed section of a source field is one of

$$\varnothing, \qquad \{a\}, \qquad [a, b],$$

and the open section is respectively empty, empty, or (a, b) unless L lies in a supporting face. The representative package stores the extreme points of the closed section in all cases and stores two off-line points when the open section is nonempty.

Let the ambient section be a closed interval $F \cap L = [c, d]$. The relative open trace is

$$(a, b) \cap [c, d],$$

which can be empty, open, half-open, or closed as a subset of L. Because the output first satisfies $L \cap \operatorname{int} P = (a, b)$ and is clipped only afterward, all four possibilities are automatic. Coincident endpoint events from different neurons are retained as literal equalities of points.

For a tangent contact $L \cap \overline{U} = \{a\}$, one adds a to the finite representative package. The final closed polygon contains a but lies on one side of the supporting line, so a is not an interior point and the strict trace is empty.

B Complete trace-case table

Let the bounded source section on a prescribed line be either empty, a point, or a compact segment, and let the ambient section be $[c, d]$. The construction retains the source section before clipping. The possibilities are summarized below; endpoint coincidences for several neurons are literal geometric coincidences and therefore remain simultaneous.

Bounded source data	Uncut strict trace	Relative strict trace	Retained finite datum
$L \cap \overline{U} = \varnothing$	$\varnothing$	$\varnothing$	none
$L \cap \overline{U} = \{a\}$	$\varnothing$	$\varnothing$	the closed point a
$L \cap \overline{U} = [a, b]$, $L \cap U = \varnothing$	$\varnothing$	$\varnothing$	the extreme points a, b
$L \cap U = (a, b)$	(a, b)	$(a, b) \cap [c, d]$	a, b and two off-line frame points

The set $(a, b) \cap [c, d]$ may be open, half-open, closed, a point, or empty. The closed relative trace is always

$$[a, b] \cap [c, d].$$

For the source-open-empty segment case, the entire closed segment is a supporting face and hence contains no ambient-interior point of the output polygon. This is distinct from a lower-dimensional neural-code atom, which may arise as the set-theoretic difference of several open fields even though each individual field is full dimensional.

Whole-line source traces. The auxiliary bounded disk is introduced before the trace package is formed. Thus an originally unbounded intersection with L becomes a compact section whose endpoints lie outside F. Freezing that bounded section preserves the entire clipped trace in F, without assigning finite endpoints to the original unbounded line section.

C Field-of-definition audit

The topological argument is over $\mathbb{R}$. The finite constraints consist of incidences, strict side relations, convex-hull containments, and exact prescribed coordinates. Three cases must be separated.

1. If the retained data are rational, the first-order existence problem has rational coefficients. The argument guarantees an output over the real algebraic numbers, but it does not in general

guarantee rational polygon vertices. The rational configurations in the verification archive are examples, not a universal rationality theorem.

2. If the retained finite data are real algebraic, real-closed-field transfer gives a real algebraic output (Basu et al., 2006).
3. If a prescribed trace endpoint or a required representative is transcendental, that coordinate is retained exactly. The natural field statement is existence in the real closure of the ordered field generated by the finite source data. No rationalization of the prescribed location is allowed.

The result is existential. The vertex bound from the unconstrained planar argument can be enlarged in terms of the finite representative package, but we do not optimize or state that bound here.

D Verification specification

The accompanying verifier uses exact rational arithmetic. A convex polygon is stored by rational vertices in counterclockwise order. Atom feasibility inside a rational ambient polygon is reduced to finitely many systems of strict and weak affine inequalities: active neurons contribute all strict facet inequalities, while an inactive neuron is handled by branching over a violated facet. Exact Fourier–Motzkin elimination determines feasibility.

The examples exercise generic crossing traces, supporting tangent contacts, closed point traces, half-open clipping, simultaneous endpoints, lower-dimensional atoms, covered carriers, shared-edge equality, and all six edge traces of a tetrahedral boundary model. These checks are regression tests for the theorem's bookkeeping. They are not a proof of the general inclusion-minimality argument.

References

Saugata Basu, Richard Pollack, and Marie-Françoise Roy. *Algorithms in Real Algebraic Geometry*, volume 10 of *Algorithms and Computation in Mathematics*. Springer, Berlin, 2 edition, 2006. doi: 10.1007/3-540-33099-2.

Boris Bukh and R. Amzi Jeffs. Planar convex codes are decidable. *SIAM Journal on Discrete Mathematics*, 37(2):951–963, 2023. doi: 10.1137/22M1511187.

Joshua Cruz, Chad Giusti, Vladimir Itskov, and Bill Kronholm. On open and closed convex codes. *Discrete & Computational Geometry*, 61:247–270, 2019. doi: 10.1007/s00454-018-9999-5.

Carina Curto, Vladimir Itskov, Alan Veliz-Cuba, and Nora Youngs. The neural ring: An algebraic tool for analyzing the intrinsic structure of neural codes. *Bulletin of Mathematical Biology*, 75: 1571–1611, 2013. doi: 10.1007/s11538-013-9860-3.

Carina Curto, Elizabeth Gross, Jack Jeffries, Katherine Morrison, Mohamed Omar, Zvi Rosen, Anne Shiu, and Nora Youngs. What makes a neural code convex? *SIAM Journal on Applied Algebra and Geometry*, 1(1):222–238, 2017. doi: 10.1137/16M1073170.

Mohammad Ghomi. Optimal smoothing for convex polytopes. *Bulletin of the London Mathematical Society*, 36(4):483–492, 2004. doi: 10.1112/S0024609303003059.

Alexander Kunin, Caitlin Lienkaemper, and Zvi Rosen. Oriented matroids and combinatorial neural codes. *Combinatorial Theory*, 3(1), 2023. doi: 10.5070/C63160427.

Rolf Schneider and John A. Wieacker. Approximation of convex bodies by polytopes. *Bulletin of the London Mathematical Society*, 13(2):149–156, 1981. doi: 10.1112/blms/13.2.149.